习题10-1 1 2(1)(3) 4(1)(3)(4) 5 6 7 8 9(2)

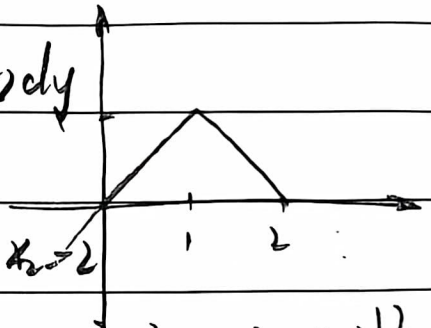
习题10-2 1(2)(3) 2(1)(2)(3) 3 4 5

习题10-3 1 2 3(1) 5(1) 6

习题10综合 4 6 7 8 12 15

八、计算下列第二类曲线积分

(1) $\int_L (x+y)dx+(x^2-y^2)dy$，其中$L$为曲线$y=1-|1-x|$ $(0\le x\le 2)$沿参数增加的方向

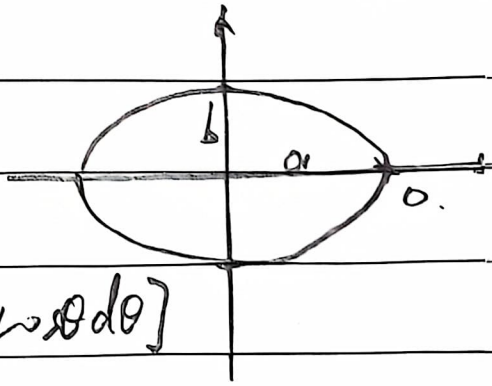


原式$=\int_{L_1}(x+y)dx+(x^2-y^2)dy+\int_{L_2}(x+y)dx+(x^2-y^2)dy$

$L_1:\begin{cases}y=x\\x=x\end{cases}$，$x_1=0$，$x_2=1$　$L_2:\begin{cases}y=2-x\\x=x\end{cases}$　$x_1=1$，$x_2=2$

$\therefore$原式$=\int_0^1 2x^2dx+\int_1^2(2x^2-8x+8)dx=\frac{2}{3}x^3\Big|_0^1+\left(\frac{2}{3}x^3-4x^2+8x\right)\Big|_1^2=$

(2) $\oint_L (x+y)dx+(x-y)dy$，其中$L$依逆时针方向通过$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



$L:\begin{cases}x=a\cos\theta\\y=b\sin\theta\end{cases}$　$\theta_1=0$，$\theta_2=2\pi$

$\therefore$原式$=\int_0^{2\pi}[-(a\cos\theta+b\sin\theta)a\sin\theta d\theta+b(a\cos\theta-b\sin\theta)\cos\theta d\theta]$

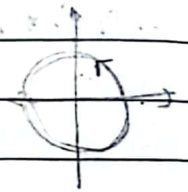
$=\int_0^{2\pi}\left[ab\cos 2\theta-\frac{1}{2}(a^2+b^2)\sin 2\theta\right]d\theta=0$

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(3) $\oint_C \frac{(x+y)dx-(x-y)dy}{x^2+y^2}$，其中C依逆时针方向通过圆周 $x^2+y^2=a^2$

原式 $=\frac{1}{a^2}\oint_C[(x+y)dx-(x-y)dy]$

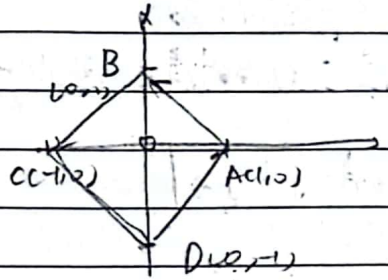
$=-\frac{2}{a^2}\iint_D d\sigma=-2\pi$



(4) $\oint_T \frac{dx+dy}{|x|+|y|}$，其中 $T=\overline{ABCDA}$ 是以A(1,0)，B(0,1)，C(-1,0)

D(0,-1)为顶点的正方形的围线

$\because |x|+|y|=1$ $\therefore$ 原式 $=\oint(0dx+0dy)=0$

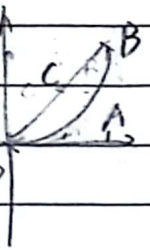



(5) $\oint_{OABCO} \arctan\frac{y}{x}dy-dx$，其中 $\overset{\frown}{OAB}$ 为抛物线段 $y=x^2$，$\overline{BCO}$ 为直线段 $y=x$，B点坐标为(1,1)

原式 $=\int_{\overset{\frown}{OAB}} \arctan\frac{y}{x}dy-dx+\int_{\overline{BCO}} \arctan\frac{y}{x}dy-dx$

$\overset{\frown}{OAB}$: $\begin{cases}y=x^2\\x=x\end{cases}$ $x_1=0$，$x_2=1$ $\overline{BCO}$: $\begin{cases}y=x\\x=x\end{cases}$ $x_1=1$ $x_2=0$

原式 $=\int_0^1(\arctan x-1)dx+\int_1^0(\frac{\pi}{4}-1)dx=\frac{\pi}{4}-1$




二、计算下列第二类曲线积分

(1) $\int_C (y^2-z^2)dx+2yzdy-x^2dz$，其中C为曲线 $x=t$，$y=t^2$，$z=t^3$ $(0\le t\le 1)$ 依参数增加的方向

原式 $=\int_0^1(t^4-t^6+2t^5\cdot 2t-t^2\cdot 3t^2)dt=\int_0^1(3t^6-2t^4)dt=\frac{1}{35}$

(3) $\int_C (y^2-z^2)dx+(z^2-x^2)dy+(x^2-y^2)dz$，其中C为球面一部分 $x^2+y^2+z^2=1$，$x\ge 0$，$y\ge 0$，$z\ge 0$ 围线，当沿着它的正方向进行时该曲面的外侧保持在左方

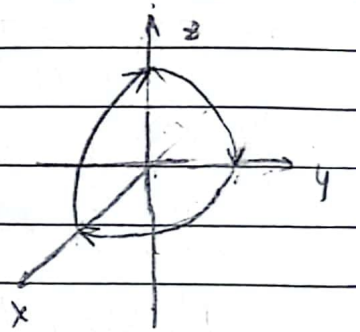


原式 $=\int_{C_1}+\int_{C_2}+\int_{C_3}$

$C_1$: $\begin{cases}x=\cos\theta\\y=\sin\theta\\z=0\end{cases}$ $\theta_1=\frac{\pi}{2}$，$\theta_2=0$

$C_2$: $\begin{cases}x=\cos\theta\\y=0\\z=\sin\theta\end{cases}$ $\theta_1=0$，$\theta_2=\frac{\pi}{2}$

$C_3$: $\begin{cases}x=0\\y=\cos\theta\\z=\sin\theta\end{cases}$ $\theta_1=\frac{\pi}{2}$，$\theta_2=0$




原式 $=\int_{\frac{\pi}{2}}^0(-\sin^3\theta-\cos^3\theta)d\theta+\int_0^{\frac{\pi}{2}}(\sin^3\theta+\cos^3\theta)d\theta+\int_{\frac{\pi}{2}}^0(-\sin^3\theta-\cos^3\theta)d\theta$

$=6\int_0^{\frac{\pi}{2}}\sin^3\theta d\theta=4$

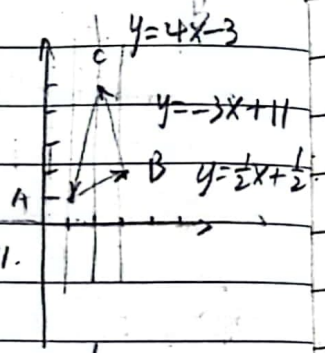
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4. 利用格林公式计算下列第二类曲线积分

(1) $\oint_C (x+y)^2dx-(x^2+y^2)dy$，其中闭路 C 依正方向依次经过以 A(1,1)，B(3,2)，C(2,5) 为顶点的三角形 ABC

原式 $=\iint_D(-\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y})d\sigma$

$=\iint_D(-4x-2y)d\sigma$




AC: $\begin{cases}\frac{1}{2}x+\frac{1}{2}\le y\le 4x-3\\ 1\le x\le 2\end{cases}$ BC: $\begin{cases}\frac{1}{2}x+\frac{1}{2}\le y\le -3x+11\\ 2\le x\le 3\end{cases}$

$\therefore$ 原式 $=\int_1^2 dx\int_{\frac{1}{2}x+\frac{1}{2}}^{4x-3}(-4x-2y)dy+\int_2^3 dx\int_{\frac{1}{2}x+\frac{1}{2}}^{-3x+11}(-4x-2y)dy=$

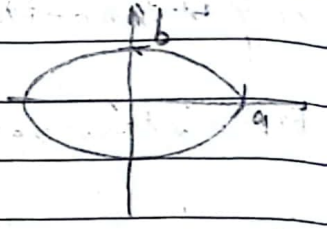
$=\int_1^2(-\frac{119}{4}x^2+\frac{77}{2}x-\frac{35}{4})dx+\int_2^3(\frac{21}{4}x^2+\frac{49}{2}x-\frac{133}{4})dx$

$=-\frac{140}{3}$

(2) $\oint_C (x+y)dx-(x-y)^2dy$，其中 C 为依正向的椭圆 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

原式 $=\iint_D(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y})d\sigma$

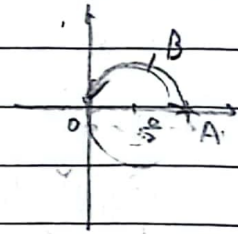
$=-2\iint_D d\sigma=-2ab$




(4) $\int_{\overset{\frown}{ABO}}(e^x\sin y-my)dx+(e^x\cos y-m)dy$，其中 $\overset{\frown}{ABO}$ 为由点 A(a,0) 至点 O(0,0) 的上半圆周 $x^2+y^2=ax$

$\frac{\partial Q}{\partial x}=e^x\cos y$ $\frac{\partial P}{\partial y}=e^x\cos y-m$

$\therefore$ 原式 $=\int_{\overset{\frown}{ABO}+\overline{OA}}+\int_{\overline{AO}}=m\iint_D d\sigma=\frac{1}{8}\pi ma^2$

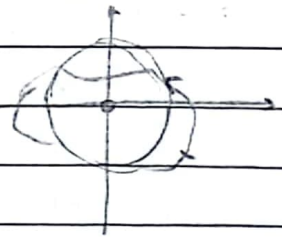



5. 计算 $I=\oint_C\frac{-ydx+xdy}{x^2+y^2}$，式中 C 为依正向而不经过坐标原点的简单封闭曲线

$\frac{\partial Q}{\partial x}=\frac{y^2-x^2}{(x^2+y^2)^2}$ $\frac{\partial P}{\partial y}=\frac{y^2-x^2}{(x^2+y^2)^2}$ $\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$

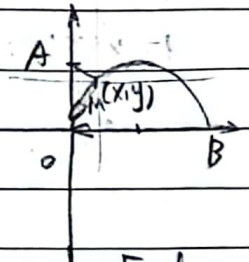
$\therefore x^2+y^2=1$，因 $I=\oint_{C_1}-ydx+xdy$

$=2\iint_D d\sigma=2\pi$



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6. 设位于点(0,1)的质点A对质点M的引力大小为$\frac{k}{r^2}$（$k>0$为万有引力常数，$r$为质点A与M的距离），质点M沿曲线$y=\sqrt{2x-x^2}$自$B(2,0)$运动到$O(0,0)$，求在此过程中质点A对质点M的引力所做的功




$r_{AM}^2 = x^2+(y-1)^2$

$F_x = \frac{Gx}{r_{AM}^3} = -\frac{Gx}{[x^2+(y-1)^2]^{\frac{3}{2}}}$

$F_y = \frac{Gy}{r_{AM}^3} = -\frac{G(y-1)}{[x^2+(y-1)^2]^{\frac{3}{2}}}$

$W = \int_C (F_x dx + F_y dy)$

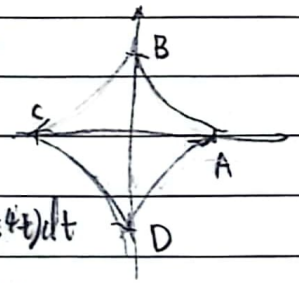
$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$ ∴ $W = \int_{\overrightarrow{BO}} F_x dx + F_y dy = \int_0^2 Gx(x^2+1)^{-\frac{3}{2}} dx$

$= G(1-\frac{\sqrt{5}}{5})$

7. 利用第二类曲线积分计算下列曲线所围的面积

(1) 星形线 $x=a\cos^3 t$, $y=b\sin^3 t$

$\frac{1}{2}\oint -y dx + x dy$?




$\iint_\sigma d\sigma = \frac{1}{2}\int_{\overrightarrow{ABCD}} (-y dx + x dy)$

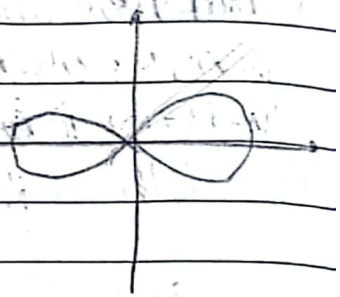
$= \frac{1}{2}\int_0^{2\pi} (3ab\cos^2 t \sin^4 t + 3ab\sin^2 t \cos^4 t) dt$

$= \frac{3}{2}ab\int_0^{2\pi} \sin^2 t \cos^2 t dt = \frac{3}{8}ab\int_0^{4\pi} \sin^2 u du = \frac{3}{8}\pi ab$

(2) 双纽线 $(x^2+y^2)^2 = a^2(x^2-y^2)$

设 $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ ⇒ ~~$r^2 = a^2\cos 2\theta$~~ ⇒ $r = a\cos^{\frac{1}{2}} 2\theta$

$\begin{cases} x = a\cos^{\frac{1}{2}} 2\theta \cos\theta \\ y = a\cos^{\frac{1}{2}} 2\theta \sin\theta \end{cases}$



$\iint d\sigma = 4\int_L (-y dx + x dy) = 2\int_0^{\frac{\pi}{4}} a^2 \cos 2\theta \, d\theta = a^2$

8 计算下列第二类曲线积分

(1) $\int_{(0,1)}^{(2,3)} (x+y)dx + (x-y)dy$

原式 $= \int_{(0,1)}^{(2,3)} d(xy + \frac{1}{2}x^2 - \frac{1}{2}y^2)$

$= 4$

(2) $\int_{(-2,-1)}^{(3,0)} (x^4+4xy^3)dx + (6x^2y^2 - 5y^4)dy$

$= \int_{(-2,-1)}^{(3,0)} d(2x^2y^2 + \frac{1}{5}x^5 - y^5)$

$= 61$

(3) $\int_{(0,1)}^{(1,0)} \frac{x dy - y dx}{(x-y)^2}$

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, 原式 $= \int_{(0,-1)}^{(0,0)} \frac{x dy - y dx}{(x-y)^2} + \int_{(0,0)}^{(1,0)} \frac{x dy - y dx}{(x-y)^2} = 0$ ?

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9、求原函数 $u$

(1) $du=(x^2+2xy-y^2)dx+(x^2-2xy-y^2)dy$

$\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$，$u=\int_{(0,0)}^{(x,0)}x^2dx+\int_{(x,0)}^{(x,y)}(x^2-2xy-y^2)dy+C$

$\Rightarrow u=\frac{1}{3}x^3+x^2y-xy^2-\frac{1}{3}y^3+C$

(2) $du=\frac{ydx-xdy}{3x^2-2xy+3y^2}$

$\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$，$u=\int_{(x,0)}^{(x,y)}\frac{-xdy}{3x^2-2xy+3y^2}=-x\int_0^y\frac{1}{3x^2-2xy+3y^2}dy$

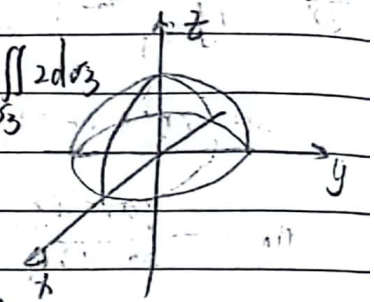
$=-x\int_0^y\frac{1}{3(y-\frac{x}{3})^2+\frac{8}{3}x^2}dy=-\frac{\sqrt{2}}{2}\int_0^y\frac{1}{(\frac{\sqrt{6}}{4}\frac{3y}{x}-1)^2+1}d(\frac{3\sqrt{2}}{4}\frac{y}{x}-\frac{\sqrt{2}}{4})$

$=-\frac{\sqrt{2}}{2}\arctan(\frac{3\sqrt{2}y}{4x}-\frac{\sqrt{2}}{4})+C$

习题10-2

1、计算下列第二类曲面积分

(2) $\iint_\Sigma yzdzdx+2dxdy$，其中$\Sigma$是球面$x^2+y^2+z^2=4$外侧在$z\geq0$的部分



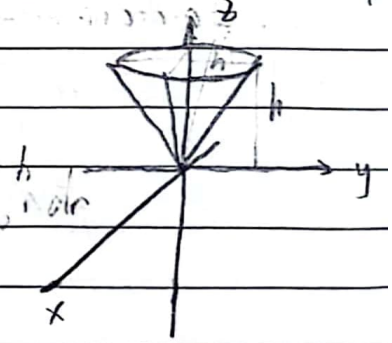

原式$=\iint_{\sigma_1}\sqrt{4-x^2-z^2}\,zd\sigma+\iint_{\sigma_2}-\sqrt{4-x^2-z^2}\,zd\sigma_2+\iint_{\sigma_3}2d\sigma_3$

设$\begin{cases}x=r\cos\theta\\z=r\sin\theta\end{cases}\Rightarrow\begin{cases}0\leq r\leq2\\0\leq\theta\leq\pi\end{cases}$

则原式$=2\int_0^\pi d\theta\int_0^2\sqrt{4-r^2}\,r\sin\theta\,rdr+8\pi=12\pi$

(3) $\iint_S(y-z)dydz+(z-x)dzdx+(x-y)dxdy$

其中$S$为圆锥曲面$x^2+y^2=z^2(0\leq z\leq h)$的外侧

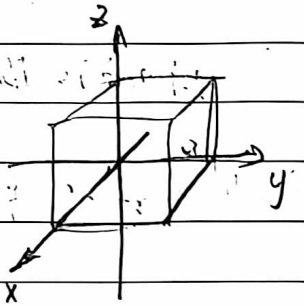



原式$=0$

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2、利用高斯公式计算下列第二类曲面积分.

(1) $\oint_S x^2dydz + y^2dzdx + z^2dxdy$，其中S为立体$0\le x\le a, 0\le y\le a, 0\le z\le a$的边界的外表面

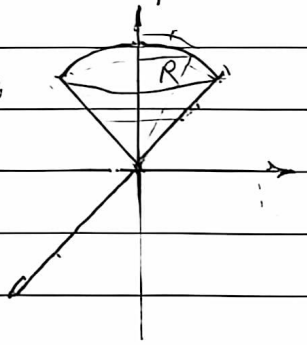



原式$=\iiint_V \left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dxdydz$

$=2\iiint_V (x+y+z)\,dxdydz$

$=2\int_0^a dx\int_0^a dy\int_0^a (x+y+z)dz = 3a^4$

(2) $\oint_S 2x^3dydz + y^2z^2dzdx - z^3dxdy$，其中S是由曲面$z=\sqrt{x^2+y^2}$与$z=\sqrt{2-x^2-y^2}$所围立体的表面外侧



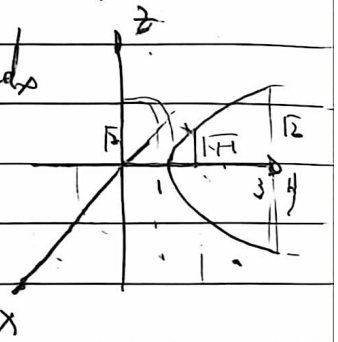

原式$=\iiint_V (6x^2+2y^2z^2-3z^2)dxdydz = \iiint_V z\,dxdydz$

$=\int_0^1 dz\iint_{D_1} z\,d\sigma_1 + \int_1^{\sqrt{2}} dz\iint_{D_2} z\,d\sigma_2$

$=\int_0^1 z\cdot z^2\,dz + \int_1^{\sqrt{2}} z\cdot\pi(2-z^2)\,dz = \frac{7}{4}$

(3) $\iint_\Sigma (8y+1)x\,dydz + 2(1-y^2)dzdx - 4yzdxdy$，其中$\Sigma$是由曲线$\begin{cases} z=\sqrt{y-1} & (1\le y\le 3) \\ x=0 \end{cases}$

绕y轴旋转一周所成的曲面，它的法向量与y轴正向夹角恒大于$\frac{\pi}{2}$




原式$=\iiint_V (8y+1+4y-4y)\,dxdydz - \iint_{S_1} 2(1-y^2)dzdx$

$=\int_1^3 \pi(y-1)dy + 3\pi = 3\pi$

3 证明：曲面$S$所包围的体积$V=\frac{1}{3}\iint_S (x\cos\alpha+y\cos\beta+z\cos\gamma)dS$，式中$\cos\alpha,\cos\beta,\cos\gamma$为曲面$S$的外法向量的方向余弦.

$\frac{1}{3}\iint_S (x\cos\alpha+y\cos\beta+z\cos\gamma)dS = \frac{1}{3}\iiint_V 3\,dV = V$

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4. 证明

$$\iint_S \left[\left(\frac{\partial R}{\partial y}-\frac{\partial Q}{\partial z}\right)\cos\alpha+\left(\frac{\partial P}{\partial z}-\frac{\partial R}{\partial x}\right)\cos\beta+\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)\cos\gamma\right]dS=0$$

其中 $S$ 是闭区域 $\Omega$ 的整个边界曲面，$P,Q,R$ 具有一阶连续偏导数

$$\text{左式}=\iiint_V\left(\frac{\partial R}{\partial y\partial x}-\frac{\partial Q}{\partial z\partial x}+\frac{\partial P}{\partial z\partial y}-\frac{\partial R}{\partial x\partial y}+\frac{\partial Q}{\partial x\partial z}-\frac{\partial P}{\partial y\partial z}\right)dv=0$$

5. 证明

(1) $\operatorname{div}(\vec{u}+\vec{v})=\operatorname{div}\vec{u}+\operatorname{div}\vec{v}$

$\vec{u}=(P,Q,R)\quad \vec{v}=(P_1,Q_1,R_1)\quad \vec{\alpha}=(\alpha_1,\alpha_2,\alpha_3)$

$\operatorname{div}(\vec{u}+\vec{v})=\frac{\partial P+P_1}{\partial x}+\frac{\partial Q+Q_1}{\partial y}+\frac{\partial R+R_1}{\partial z}=\operatorname{div}\vec{u}+\operatorname{div}\vec{v}$

(2) $\operatorname{div}(u\vec{v})=u\operatorname{div}\vec{v}$

$\operatorname{div}(u\vec{v})=\frac{\partial uP}{\partial x}+\frac{\partial uQ}{\partial y}+\frac{\partial uR}{\partial z}=u\operatorname{div}\vec{v}$

$+v\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+v\frac{\partial u}{\partial z}$

(3) $\operatorname{div}(\vec{\alpha}u)=\vec{\alpha}\cdot\operatorname{grad}u$

$\operatorname{div}\vec{\alpha}u=\frac{\partial(\alpha_1 P)}{\partial x}+\frac{\partial(\alpha_2 Q)}{\partial y}+\frac{\partial(\alpha_3 R)}{\partial z}=(\alpha_1,\alpha_2,\alpha_3)\left(\frac{\partial P}{\partial x},\frac{\partial Q}{\partial y},\frac{\partial R}{\partial z}\right)$

$=\vec{\alpha}\cdot\operatorname{grad}u$

六题1-3

1. 分别利用格点积和向量计算[1]第二类曲线积分

$\oint_L (yz+z)dx+(zx+x)dy+(xy+y)dz,$

$L$ 为 $C$ 为以椭球面与平面 $x+y=1$ 的交线为界的闭

$x=\cos t,\ y=\sin t\cos t,\ z=\sin t\cos t \quad (0\le t\le 2\pi)$




$$\begin{cases} y=xz \\ x+z=a \end{cases}$$

原式 $=\iint_S\left(\frac{\partial R}{\partial y}-\frac{\partial Q}{\partial z}\right)dydz+\left(\frac{\partial P}{\partial z}-\frac{\partial R}{\partial x}\right)dzdx$

$+\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=0$

2. 设 $C$ 为以下闭圆 $x\cos\alpha+y\cos\beta+z\cos\gamma-p=0$ 与 $S$ 闭圆锥面

其中圆锥面 $S$ 是依正方向进行的，证明 $\oint_C$ 的值与 $S$ 的形状无关

$$\oint_C (z\cos\beta-y\cos\gamma)dx+(x\cos\gamma-z\cos\alpha)dy+(y\cos\alpha-x\cos\beta)dz$$

$$\text{左式}=\iint_S\cos\alpha dydz+\cos\beta dzdx+\cos\gamma dxdy=2S$$

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3、计算下列第二类曲线积分

(1) $\int_{(1,1,1)}^{(2,3,4)} yz\,dx + xz\,dy + xy\,dz$

$= xyz\Big|_{(1,1,1)}^{(2,3,4)} = 0$

5、求下列函数的原函数

(1) $du = (x^2 - yz)dx + (y^2 - 2xz)dy + (z^2 - 2xy)dz$

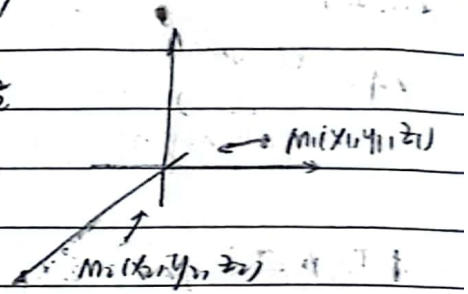
$du = x^2dx + y^2dy + z^2dz - 2(yz\,dx + xz\,dy + xy\,dz)$

$= d(\frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - 2xyz)$

$\therefore u = \frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - 2xyz + C$

6、当单位质量从点 $M_1(x_1,y_1,z_1)$ 移动到点 $M_2(x_2,y_2,z_2)$ 时，作用于单位质量的引力 $F$ 的大小为 $|F| = \frac{G}{r^2}$，$r = \sqrt{x^2+y^2+z^2}$，方向指向原点 O，求引力所做的功

$r_1 = \sqrt{x_1^2+y_1^2+z_1^2}$　　$r_2 = \sqrt{x_2^2+y_2^2+z_2^2}$




$W = -\int_{r_1}^{r_2} F\,dr = GM(\frac{1}{r_1} - \frac{1}{r_2})$

$= GM(\frac{1}{\sqrt{x_2^2+y_2^2+z_2^2}} - \frac{1}{\sqrt{x_1^2+y_1^2+z_1^2}})$

第十章综合题

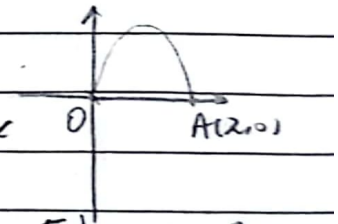
4. 在过点 O(0,0) 和 A(π,0) 的曲线族 $y = a\sin x$ ($a > 0$) 中，求一条曲线 L，使沿该曲线从 O 到 A 的积分 $\int_L (1+y^3)dx + (2x+y)dy$ 的值最小

$\begin{cases} y = a\sin x \\ x = x \end{cases}$　$x_1 = 0,\ x_2 = \pi$

原式 $= \int_0^{\pi} (1 + a^3\sin^3 x)dx + (2x + a\sin x)a\cos x\,dx$

$= \frac{4}{3}a^3 - 4a + 2$

$f(a) = \frac{4}{3}a^3 - 4a$　$f'(a) = 4a^2 - 4$




$a = 1$ 时原式有极小值

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6、设$f(x)$具有一阶连续导数

证明对任何光滑闭曲线$L$，有$\oint_L f(xy)(y\,dx+x\,dy)=0$

设$M=xy$，$\oint_L f(xy)\,dx+xf(xy)\,dy$

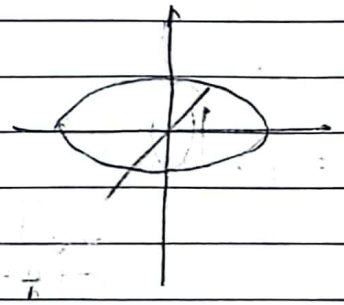
$$=f(xy)+xy\frac{\partial f}{\partial M}-[f(xy)+yx\frac{\partial f}{\partial M}]=0$$

7、在变力$\vec{F}=yz\vec{i}+zx\vec{j}+xy\vec{k}$作用下

质点从原点沿直线运动到椭球面$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

上第一卦限点$M(\xi,\eta,\zeta)$，$\xi,\eta,\zeta$何值时力$\vec{F}$做功$W$最大

求出$W$的最大值



$OM:\ \frac{x}{\xi}=\frac{y}{\eta}=\frac{z}{\zeta}=k\Rightarrow\begin{cases}x=\xi k\\y=\eta k\\z=\zeta k\end{cases}$

$W=\int_L \vec{F}\cdot d\vec{s}\qquad k_1=0,\ k_2=1$

$=\int_L yz\,dx+zx\,dy+xy\,dz$

$=3\xi\eta\zeta\int_{k_1}^{k_2}k^2\,dk=\xi\eta\zeta$

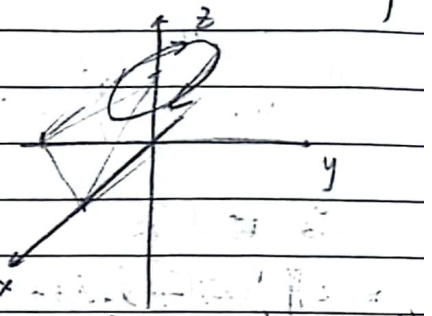
$\frac{\xi^2}{a^2}+\frac{\eta^2}{b^2}+\frac{\zeta^2}{c^2}\geq 3\sqrt[3]{\frac{\xi^2\eta^2\zeta^2}{a^2b^2c^2}}\Rightarrow$

当$\xi=\frac{\sqrt{3}}{3}a,\ \eta=\frac{\sqrt{3}}{3}b,\ \zeta=\frac{\sqrt{3}}{3}c$时

$W$有最大值$\frac{\sqrt{3}}{9}abc$

8、计算$\oint_L (z-y)\,dx+(x-z)\,dy+(x-y)\,dz$，

其中$L$是曲线$\begin{cases}x^2+y^2=1\\x-y+z=2\end{cases}$，从$Oz$轴正向看去$L$方向顺时针




设$\begin{cases}x=\cos\theta\\y=\sin\theta\\z=2+\sin\theta-\cos\theta\end{cases}$　$\theta_1=0,\ \theta_2=-2\pi$

原式$=\int_0^{-2\pi}-(2-\cos\theta)\sin\theta+(2\cos\theta-\sin\theta-2)\cos\theta+(\cos\theta-\sin\theta)(\cos\theta+\sin\theta)\,d\theta$

$=\int_0^{-2\pi}(3\cos^2\theta-2\cos\theta-\sin^2\theta-2\sin\theta)\,d\theta$

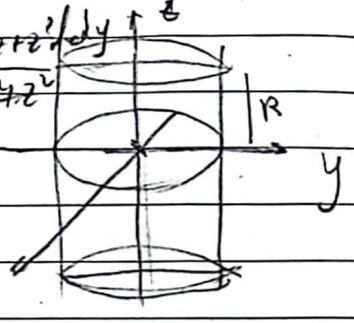
$=\int_0^{-2\pi}2\sin^2\theta\,d\theta=2\pi$

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12 计算曲面积分 $\iint_S \frac{x\,dy\,dz+z^2\,dx\,dy}{x^2+y^2+z^2}$，其中S是曲面$x^2+y^2=R^2$，

与两平面$z=R$，$z=-R(R>0)$所围成立体表面的外侧

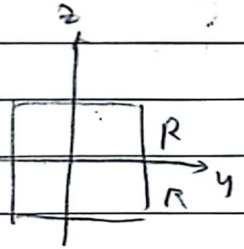
原式 $=\iint_{S_1}\frac{z^2\,dx\,dy}{x^2+y^2+z^2}+\iint_{S_2}\frac{z^2\,dx\,dy}{x^2+y^2+z^2}+\iint_{S_3}\frac{x\,dy\,dz+z^2\,dx\,dy}{x^2+y^2+z^2}$




$=\iint_{S_3}\frac{x\,dy\,dz}{x^2+y^2+z^2}$

$=2\iint_{\sigma_1}\frac{\sqrt{R^2-y^2}}{R^2+z^2}\,dy\,dz$

$=2\int_{-R}^{R}dy\int_{-R}^{R}\frac{\sqrt{R^2-y^2}}{R^2+z^2}\,dz$




$=2R^2\int_{-R}^{R}\frac{1}{R^2+z^2}\,dz=\frac{1}{2}\pi^2R$



15 计算 $I=\oint_S\frac{\cos(\vec{r},\vec{n})}{r^2}\,ds$ 其中 $\vec{r}=x\vec{i}+y\vec{j}+z\vec{k}$

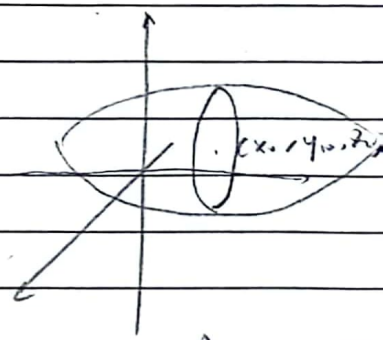
$r=\sqrt{x^2+y^2+z^2}$

S为不经过原点的椭球面 $\frac{(x-x_0)^2}{a^2}+\frac{(y-y_0)^2}{b^2}+\frac{(z-z_0)^2}{c^2}=1$

$\vec{n}$为S的单位外法矢量

$I=\oint_S\frac{\vec{r}\cdot\vec{n}\,ds}{r^3}=\oint_S\frac{x\,dy\,dz+y\,dz\,dx+z\,dx\,dy}{(x^2+y^2+z^2)^{\frac{3}{2}}}$

$\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=0$

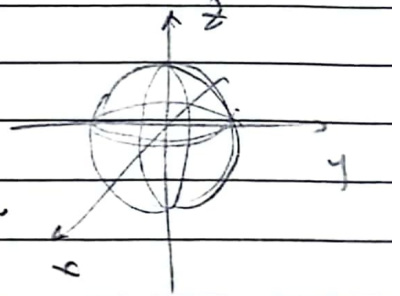



若O不在S内部，$I=0$

若O在S内部

对 $x^2+y^2+z^2=1$

$I=\oint_S x\,dy\,dz+y\,dz\,dx+z\,dx\,dy$




$=6\iint_{\sigma_1}d\sigma_1=\int_0^{2\pi}d\theta\int_0^1\sqrt{1-r^2}\,r\,dr=4\pi$

$=$

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